

Boundaried Monoids via Comprehension

A purely equational theory

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Abstract

We give a self-contained, purely equational theory $\mathbf{T}$ of what we call *boundaried monoids*: a generalisation of monoids in which every element carries a left and a right boundary drawn from an auxiliary sort, and the product of two elements is defined exactly when the inner boundaries agree. The development uses no notion beyond sorts, operations, equations, and *comprehension*. Its one structural idea is presentational: rather than treat the product as a partial operation hedged by side conditions, we absorb its partiality into a *comprehension structure*, so that the product becomes a total operation valued in a collection sort. The translation between the partial and the total readings is exactly comprehension in the sense of Wadler [1]. We axiomatise the comprehension layer (§2), state the principal theory over it (§3), give an equivalent dependently sorted presentation, and prove the two equivalent with comprehension as the translating device (§4). Throughout, the object of study is an algebra and nothing more.

1 Aim and standing conventions

This note isolates a single algebraic theory and writes down its equations. We deliberately confine ourselves to elementary algebraic vocabulary: a *signature* is a family of sorts together with operations between finite products of sorts; a *theory* adds equations; a *model* is a sort-indexed family of carriers interpreting the operations and satisfying the equations. Beyond this we use only one further device, *comprehension*, which we axiomatise in §2 before it is used.

The structure we axiomatise is a mild generalisation of a monoid. Recall that a monoid is a set with an associative binary product and a two-sided neutral element. A *boundaried monoid* relaxes totality of the product: each element f is assigned a left boundary $\partial_0 f$ and a right boundary $\partial_1 f$ in an auxiliary sort $\mathbf{O}$, and the product $g \cdot f$ is intended to exist precisely when $\partial_0 g = \partial_1 f$. A monoid is the special case in which $\mathbf{O}$ has a single element, so that every pair is multipliable.

There are two faithful ways to render the partial product. The first keeps the product partial and indexes it by boundaries, giving a dependently sorted *equational* theory with totally standard operations (§4). The second keeps a single sort of elements and makes the product *total* by valuing it in a collection sort: the product of a non-multipliable pair is the empty collection, and the product of a multipliable pair is a singleton. The second reading is the one in which the partiality is handled by comprehension, and it is the reading we take as primary. The bulk of the note shows that the two readings define the same theory.

A note on scope. We make no semantic claims here; interpretations of $\mathbf{T}$ are deferred to separate work. We also make no use of any structure richer than the comprehension layer of §2. The reader will recognise that layer, under a different packaging, as a familiar gadget; we cite its provenance [1, 2] and otherwise do not rely on that packaging.¹

¹The comprehension structure of Definition 2.1 is an *extension system*: a unit together with a Kleisli-style extension subject to three equations. We present it in exactly this elementary form so that the present theory

2 The comprehension layer

We work over a fixed many-sorted signature that is closed under finite products: it has a terminal sort $\mathbf{1}$ with unique element $*$, binary products $X \times Y$ with projections and pairing $\langle, \rangle$, and the evident equations. We write $f: X \rightarrow Y$ for a derived operation (a term in one free variable). All of this is ordinary equational algebra.

Definition 2.1 (Comprehension structure). A *comprehension structure* consists of

- (i) a sort MX for every sort X (the sort of *collections* of X);
- (ii) for each sort X a *unit* $\eta_X: X \rightarrow MX$;
- (iii) for each operation $f: X \rightarrow MY$ an *extension* $f^*: MX \rightarrow MY$;
- (iv) for each sort X a constant $\emptyset_X: MX$ (the *empty collection*);
- (v) an *equality guard* $\langle - \doteq - \rangle: \mathbf{O} \times \mathbf{O} \rightarrow M\mathbf{1}$ on the distinguished sort $\mathbf{O}$, subject to the equations

$$\eta_X^* = \text{id}_{MX}, \quad (\text{C1})$$

$$f^* \circ \eta_X = f, \quad (\text{C2})$$

$$(g^* \circ f)^* = g^* \circ f^*, \quad (\text{C3})$$

for all $f: X \rightarrow MY$, $g: Y \rightarrow MZ$; the *commutativity* law

$$[(x, y) \mid x \leftarrow u, y \leftarrow v] = [(x, y) \mid y \leftarrow v, x \leftarrow u] \quad (\text{C4})$$

whenever $u: MX$ and $v: MY$ share no free variable; the *zero* laws

$$f^* \circ \emptyset_X = \emptyset_Y, \quad (\text{C5})$$

$$[t \mid x \leftarrow \emptyset_X, \bar{q}] = \emptyset, \quad (\text{C6})$$

and the *guard* laws

$$\langle a \doteq a \rangle = \eta_{\mathbf{1}}(*), \quad (\text{C7})$$

$$[\varphi(a, b) \mid \langle a \doteq b \rangle, \bar{q}] = [\varphi(a, a) \mid \langle a \doteq b \rangle, \bar{q}]. \quad (\text{C8})$$

The comprehension notation $[\cdot \mid \cdot]$ used above is defined in Notation 2.2; equations (C4)–(C8) are to be read after that expansion.

Notation 2.2 (Comprehension). A *qualifier* is either a *generator* $x \leftarrow u$ (with $u: MX$ and x a fresh variable of sort X) or a *guard* $\langle a \doteq b \rangle$ (with $a, b: \mathbf{O}$). For a term t and a finite list $\bar{q} = q_1, \dots, q_n$ of qualifiers we define a term $[t \mid \bar{q}]: MY$ (where $t: Y$ in the context bound by $\bar{q}$) by recursion on the list:

$$\begin{aligned} [t \mid] &= \eta(t), \\ [t \mid x \leftarrow u, \bar{q}] &= (\lambda x. [t \mid \bar{q}])^* u, \\ [t \mid \langle a \doteq b \rangle, \bar{q}] &= (\lambda *. [t \mid \bar{q}])^* \langle a \doteq b \rangle. \end{aligned}$$

Thus generators are extension, the head is the unit, and guards filter. We read $[t \mid \bar{q}]$ as “collect t as the bound variables range over the generators, retaining only those bindings that satisfy the guards.”

remains an algebra, with no appeal to functoriality or to ambient morphisms.

The equations have transparent readings. (C1)–(C3) say that collecting a singleton is identity, that a singleton generator binds by substitution, and that nested collection may be flattened in either order: they are the laws that make comprehension well defined. (C4) says independent generators may be reordered. (C5)–(C6) say that nothing is collected over an empty range, from either side. (C7) says a reflexive guard is vacuous (it contributes the singleton $\eta(*)$, which passes through); (C8) says that under a guard $\langle a \doteq b \rangle$ one may rewrite b to a . We record the two immediate consequences we shall need.

Lemma 2.3 (Reduction). *For all suitable terms,*

$$[t \mid \langle a \doteq a \rangle, \bar{q}] = [t \mid \bar{q}], \quad (\text{R1})$$

$$[t \mid x \leftarrow \eta(s), \bar{q}] = [t[s/x] \mid \bar{q}]. \quad (\text{R2})$$

Proof. (R1): by Notation 2.2 the left side is $(\lambda * . [t \mid \bar{q}])^* \langle a \doteq a \rangle$, which by (C7) equals $(\lambda * . [t \mid \bar{q}])^* (\eta *)$, and this is $[t \mid \bar{q}]$ by (C2). (R2): the left side is $(\lambda x . [t \mid \bar{q}])^* (\eta s)$, which is $[t \mid \bar{q}][s/x] = [t[s/x] \mid \bar{q}]$ by (C2). $\square$

Notation 2.4 (Subsort by comprehension). For a sort X and parallel terms $p, q: X \rightarrow \mathbf{O}$ we write

$$\{x: X \mid p(x) \doteq q(x)\}$$

for the carrier whose elements are those $x: X$ with $p(x) = q(x)$, with the inclusion term and the universal property of an equationally defined subsort. Under the comprehension reading this subsort is the support of $[x \mid x \leftarrow w, \langle p(x) \doteq q(x) \rangle]$ for w ranging over MX ; the guard discards the elements failing the equation, by (C6)–(C8). We use the subsort notation freely below as a convenient name for this support.

3 The theory $\mathbf{T}$ of boundaried monoids

We now state the principal theory over a fixed comprehension structure.

Definition 3.1 (Signature). The signature of $\mathbf{T}$ extends that of §2 with

- (i) two sorts $\mathbf{O}$ (*boundaries*) and $\mathbf{E}$ (*elements*), with $\mathbf{O}$ the distinguished sort carrying the guard of Definition 2.1;
- (ii) *boundary* operations $\partial_0, \partial_1: \mathbf{E} \rightarrow \mathbf{O}$;
- (iii) a *neutral* operation $\mathbf{u}: \mathbf{O} \rightarrow \mathbf{E}$;
- (iv) a *multiplication* $\kappa: \mathbf{E} \times \mathbf{E} \rightarrow M\mathbf{E}$.

We read $\kappa(g, f)$ as “the collection of products of g after f ,” a total operation whose value is empty when f, g are not multipliable and a singleton otherwise.

Definition 3.2 (Derived collections). Over the signature of Definition 3.1 we name three comprehensions.

$$\begin{aligned} \mathbf{H}(a, b) &:= \{f: \mathbf{E} \mid \partial_0 f \doteq a, \partial_1 f \doteq b\} && \text{(elements with prescribed boundaries),} \\ \mathbf{E}^{(2)} &:= \{(g, f): \mathbf{E} \times \mathbf{E} \mid \partial_0 g \doteq \partial_1 f\} && \text{(multipliable pairs),} \\ \mathbf{E}^{(3)} &:= \{(h, g, f) \mid \partial_0 h \doteq \partial_1 g, \partial_0 g \doteq \partial_1 f\} && \text{(multipliable triples).} \end{aligned}$$

Each is a subsort in the sense of Notation 2.4; none is a new primitive. The multiplication κ is defined on all of $\mathbf{E} \times \mathbf{E}$, but the axioms below force it to be supported on $\mathbf{E}^{(2)}$.

Definition 3.3 (Axioms of $\mathbf{T}$). A model of $\mathbf{T}$ is a comprehension structure together with interpretations of $\mathbf{O}, \mathbf{E}, \partial_0, \partial_1, \mathbf{u}, \kappa$ satisfying the following equations, for all $a: \mathbf{O}$ and $f, g, h: \mathbf{E}$.

Boundaries of the neutral element.

$$\partial_0(\mathbf{u}(a)) = a, \quad (\text{T1})$$

$$\partial_1(\mathbf{u}(a)) = a. \quad (\text{T2})$$

Support and determinacy of the product.

$$[* \mid v \leftarrow \kappa(g, f)] = \langle \partial_0 g \doteq \partial_1 f \rangle, \quad (\text{T3})$$

$$[(v, v') \mid v \leftarrow \kappa(g, f), v' \leftarrow \kappa(g, f)] = [(v, v) \mid v \leftarrow \kappa(g, f)]. \quad (\text{T4})$$

Boundaries of a product.

$$[\partial_0 v \mid v \leftarrow \kappa(g, f)] = [\partial_0 f \mid v \leftarrow \kappa(g, f)], \quad (\text{T5})$$

$$[\partial_1 v \mid v \leftarrow \kappa(g, f)] = [\partial_1 g \mid v \leftarrow \kappa(g, f)]. \quad (\text{T6})$$

Neutrality.

$$\kappa(\mathbf{u}(\partial_1 f), f) = \eta(f), \quad (\text{T7})$$

$$\kappa(f, \mathbf{u}(\partial_0 f)) = \eta(f). \quad (\text{T8})$$

Associativity.

$$[w \mid v \leftarrow \kappa(g, f), w \leftarrow \kappa(h, v)] = [w \mid v \leftarrow \kappa(h, g), w \leftarrow \kappa(v, f)]. \quad (\text{T9})$$

A few words on why these nine equations say what is intended.

(T3) is the *support* law: mapping the collection $\kappa(g, f)$ onto the terminal sort yields $\langle \partial_0 g \doteq \partial_1 f \rangle$, which is the singleton $\eta(*)$ when g, f are multipliable and is empty otherwise (by (C7) and the discussion of Notation 2.4). Hence $\kappa(g, f)$ is non-empty exactly on $\mathbf{E}^{(2)}$.

(T4) is *determinacy*: any two elements drawn from $\kappa(g, f)$ coincide, so $\kappa(g, f)$ has at most one element. With (T3) this makes $\kappa(g, f)$ a singleton on $\mathbf{E}^{(2)}$ and empty off it. Partiality has been wholly absorbed into the collection sort; κ itself is total.

(T5)–(T6) transport boundaries across a product: wherever a product exists, its left boundary is that of f and its right boundary is that of g . Stating these by mapping over $\kappa(g, f)$ keeps them equational and vacuously true off $\mathbf{E}^{(2)}$.

(T7)–(T8) are the two neutral laws. The pair on the left of (T7) is multipliable since $\partial_0(\mathbf{u}(\partial_1 f)) = \partial_1 f$ by (T1), so the guard in (T3) passes and the law asserts that the product is the singleton $\eta(f)$; symmetrically for (T8).

(T9) is associativity. Both sides are double comprehensions: the left forms g after f , then h after that; the right forms h after g , then that after f . By (T3) each side is non-empty exactly when $\partial_0 g \doteq \partial_1 f$ and $\partial_0 h \doteq \partial_1 g$, the common condition defining $\mathbf{E}^{(3)}$, and on that subsort each is a singleton; the equation asserts the two singletons agree. Note that the boundary bookkeeping that ordinarily accompanies associativity is here discharged automatically by the guards inside κ , via (T3) and (T5)–(T6).

Remark 3.4 (Monoids as a special case). If $\mathbf{O} \cong \mathbf{1}$ then ∂_0, ∂_1 are constant, every pair is multipliable, (T3) makes $\kappa(g, f)$ a singleton for all g, f , and (T7)–(T9) reduce to the ordinary unit and associativity laws of a monoid with unit $\mathbf{u}(*)$. Thus $\mathbf{T}$ specialises to the equational theory of monoids.

Lemma 3.5 (Single-valued product). *In any model of $\mathbf{T}$ there is, for each multipliable pair $(g, f) \in \mathbf{E}^{(2)}$, a unique element $g \bullet f: \mathbf{E}$ with $\kappa(g, f) = \eta(g \bullet f)$, and it satisfies $\partial_0(g \bullet f) = \partial_0 f$, $\partial_1(g \bullet f) = \partial_1 g$.*

Proof. On $E^{(2)}$ the guard $\langle \partial_0 g \doteq \partial_1 f \rangle$ equals $\eta(*)$ by (C7), so (T3) gives $[* \mid v \leftarrow \kappa(g, f)] = \eta(*)$: the collection $\kappa(g, f)$ is non-empty. By (T4) it is a subsingleton. A non-empty subsingleton is a singleton $\eta(g \bullet f)$ for a unique $g \bullet f$; uniqueness is by (C2). The boundary identities follow from (T5)–(T6) and (R2) applied to $\kappa(g, f) = \eta(g \bullet f)$. $\square$

4 The fibred presentation, and equivalence

We now give the second, dependently sorted reading and show it defines the same theory. Here the product is honestly total and single-valued, and no zero or guard is needed; the price is a sort that depends on two parameters.

Definition 4.1 (The fibred theory $\mathbf{T}^b$). The theory $\mathbf{T}^b$ has

- (i) a sort $\mathbf{O}$, and for each $a, b: \mathbf{O}$ a sort $\mathbf{H}(a, b)$;
 - (ii) neutral elements $\mathbf{e}_a: \mathbf{H}(a, a)$ for each $a: \mathbf{O}$;
 - (iii) products $\diamond_{a,b,c}: \mathbf{H}(b, c) \times \mathbf{H}(a, b) \rightarrow \mathbf{H}(a, c)$ for each $a, b, c: \mathbf{O}$;
- subject to the equations, for $f: \mathbf{H}(a, b)$, $g: \mathbf{H}(b, c)$, $h: \mathbf{H}(c, d)$,

$$\mathbf{e}_b \diamond f = f, \tag{F1}$$

$$f \diamond \mathbf{e}_a = f, \tag{F2}$$

$$h \diamond (g \diamond f) = (h \diamond g) \diamond f. \tag{F3}$$

This is an ordinary equational theory, with total single-valued operations and three equations; it uses no comprehension structure.

Proposition 4.2 (Equivalence). *$\mathbf{T}$ and $\mathbf{T}^b$ are definitionally equivalent: there are mutually inverse interpretations of each theory in the other, definable using the comprehension structure, that are the identity on $\mathbf{O}$.*

Proof sketch. From $\mathbf{T}$ to $\mathbf{T}^b$. Keep $\mathbf{O}$. Define $\mathbf{H}(a, b)$ by the comprehension of Definition 3.2. Put $\mathbf{e}_a := u(a)$, which lies in $\mathbf{H}(a, a)$ by (T1)–(T2). For $g \in \mathbf{H}(b, c)$ and $f \in \mathbf{H}(a, b)$ the pair (g, f) is multipliable, so Lemma 3.5 supplies a unique $g \bullet f$ with $\kappa(g, f) = \eta(g \bullet f)$, lying in $\mathbf{H}(a, c)$ by the boundary identities; set $g \diamond f := g \bullet f$. Equations (F1)–(F2) are (T7)–(T8) with the unit stripped by (C2). For (F3): apply (R2) twice inside each side of (T9), replacing $\kappa(g, f)$ by $\eta(g \bullet f)$ and then $\kappa(h, g \bullet f)$ by $\eta(h \bullet (g \bullet f))$ on the left, and symmetrically on the right; (T9) then reads $\eta(h \bullet (g \bullet f)) = \eta((h \bullet g) \bullet f)$, whence (F3) by (C2).

From $\mathbf{T}^b$ to $\mathbf{T}$. Keep $\mathbf{O}$. Define the sort of elements as the comprehension (disjoint collection)

$$\mathbf{E} := \{ (a, b, f) \mid a, b: \mathbf{O}, f: \mathbf{H}(a, b) \},$$

with $\partial_0(a, b, f) := a$, $\partial_1(a, b, f) := b$, and $u(a) := (a, a, \mathbf{e}_a)$. Define the multiplication as the guarded singleton

$$\kappa((b', c, g), (a, b, f)) := [(a, c, g \diamond f) \mid \langle b' \doteq b \rangle],$$

which by (C7) is $\eta(a, c, g \diamond f)$ when $b' = b$ and is $\emptyset$ otherwise (the $g \diamond f$ on the right is well typed precisely under the guard, after rewriting b' to b by (C8)). One checks the axioms directly: (T3) and (T4) hold because the value is a guarded singleton; (T5)–(T6) because the boundaries of $(a, c, g \diamond f)$ are a and c ; (T7)–(T8) from (F1)–(F2); and (T9) from (F3), the two guards on each side combining to the single condition defining $\mathbf{E}^{(3)}$.

Mutual inverse. Starting from $\mathbf{T}$, forming $\mathbf{T}^b$ and then $\mathbf{T}$ again returns $\mathbf{E}$ up to the evident identification $f \mapsto (\partial_0 f, \partial_1 f, f)$, under which $\partial_0, \partial_1, u, \kappa$ are preserved on the nose by Lemma 3.5. Starting from $\mathbf{T}^b$, the round trip returns $\mathbf{H}(a, b)$ as the comprehension $\{ (a', b', f) \mid a' \doteq a, b' \doteq b \}$, naturally identified with $\mathbf{H}(a, b)$ by (C8). Both identifications respect all operations, so the interpretations are mutually inverse. The only non-equational devices used in either direction are comprehensions, which is the asserted translating role of comprehension. $\square$

Corollary 4.3. *$\mathbf{T}$ is conservative over the comprehension layer of §2: any equation between terms not mentioning $\partial_0, \partial_1, \mathbf{u}, \kappa$ that holds in $\mathbf{T}$ already holds in the comprehension layer. In particular adding the boundaried-monoid operations introduces no new collection identities.*

Proof. By Proposition 4.2 it suffices to argue for $\mathbf{T}^b$, whose operations are disjoint in sort from those of the comprehension layer and whose axioms (F1)–(F3) involve no comprehension former; hence they cannot entail a new identity among comprehension terms. $\square$

5 Remarks and scope

Free models and chains. The free boundaried monoid on a family of generating elements with prescribed boundaries has, as its elements over a pair (a, b) , the finite multipliable chains from a to b , with the empty chain at each a as neutral element and concatenation as product. The passage from generators to chains is itself a comprehension: a chain is collected by iterating the multipliability guard along its links, exactly the list-forming instance of the comprehension calculus of §2. Thus both the carving of multipliable pairs and the generation of free elements are handled by the same device.

Determinacy is optional. Dropping (T4) yields a strictly weaker theory in which $\kappa(g, f)$ may be a genuine collection rather than a subsingleton: a “multivalued” boundaried monoid, in which several products of a multipliable pair coexist. The remaining axioms still govern boundaries, support, neutrality, and associativity. We record this only to note that (T4) is the single axiom responsible for single-valuedness; it is the natural knob to turn when one later wishes to relax it.

What the comprehension structure may vary. Nothing in §3–§4 fixes *which* comprehension structure underlies M , beyond the equations (C1)–(C8). Varying that structure—while keeping the equations (T1)–(T9) verbatim—yields a family of theories sharing the present axiomatics. The systematic study of that family, and of the interpretations it admits, is deferred; here we have only fixed the equations and shown the partial and total presentations to be one theory.

Summary. The theory $\mathbf{T}$ of Definition 3.3 is a purely equational, finitely axiomatised algebra. Its single design choice is to absorb the partiality of a boundaried product into a comprehension structure, making the product a total operation valued in a collection sort. Proposition 4.2 shows this total presentation and the dependently sorted partial presentation to be definitionally equivalent, with comprehension as the explicit translation between them.

References

- [1] P. Wadler. *Comprehending monads*. Mathematical Structures in Computer Science **2**(4):461–493, 1992.
- [2] E. Moggi. *Notions of computation and monads*. Information and Computation **93**(1):55–92, 1991.